

AEROSPACE ENGINEERING

Nathan Xia

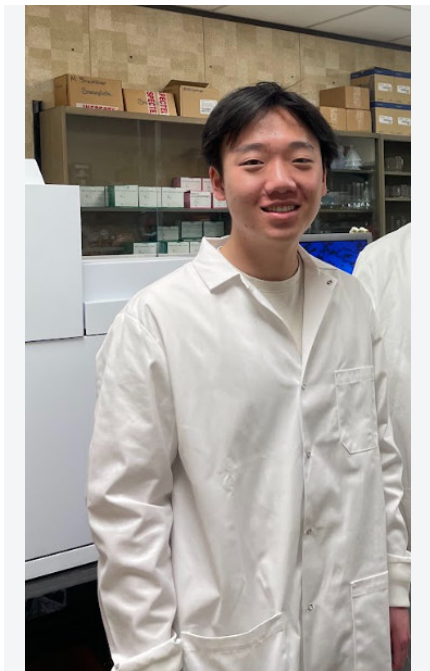
Second-Year Engineering Science Student | University of Toronto

Flow control, aeroacoustics, and experimental & computational fluid dynamics.

ABOUT ME

I’m a second-year Engineering Science student at the University of Toronto, intending to specialize in Aerospace Engineering. My interests centre on flow control and experimental and computational fluid dynamics, with the goal of pursuing graduate research and aerospace R&D.

My work spans aeroacoustic calibration research at UTIAS, aerodynamic design and mechanical leadership with RSX Aerial, and development of an interactive CFD solver. Across these projects, I combine simulation, experimental design, and hands-on manufacturing to investigate flow behaviour and improve engineering designs.



University of Calgary Research
Research opportunity in the Applied Geochemistry Research Group.

VIEW PROJECTS

DOWNLOAD RESUME

VIEW WEBSITE

TOOLS AND LANGUAGES

CAD & Manufacturing	Fusion 360 · SolidWorks
Simulation	ANSYS Fluent · Autodesk CFD · OpenVSP / VSPAERO
Programming & Controls	Python · MATLAB · C · Java · Arduino IDE

SELECTED PROJECTS

01	Aeroacoustics Parametric Research Project	2-4
02	CanSat Autonomous Glider	5-6
03	Flow Control Research Project	7-8
04	Interactive 2D CFD Solver	9-10
05	Vortex Generators: Fluent CFD Study	11-13
06	Tabletop Wind Tunnel	14-15

nathanxia123@gmail.com

(825) 994-0165 | Toronto, ON

LinkedIn: Nathan Xia

01 / UTIAS RESEARCH

Aeroacoustics Parametric Research Project

Undergraduate Research Intern | Summer 2026

The Problem

RMP calibration changes with probe geometry and test conditions, introducing uncertainty into aeroacoustic measurements.

Research Scope

- Designed the apparatus and test matrix for a 16-variable parametric study.
- Built a MATLAB acquisition and analysis pipeline for transfer functions and repeatability.
- Developed printed components and a sensor circuit retained by the research group.



Environmental Sensor Unit

Custom enclosure and display for ambient-condition measurements.

69% lower TF magnitude standard deviation with the revised calibration apparatus, relative to the control.

EXPERIMENTAL CONTEXT

At UTIAS, I independently led a 16-variable study of remote microphone probe (RMP) calibration in Professor Philippe Lavoie’s Flow Control and Experimental Turbulence (FCET) Laboratory. I was one of two first-year students conducting research at UTIAS.

RMPs connect a surface pressure inlet to a remotely located microphone through tubing. Propagation through the tube attenuates the signal and shifts its phase, requiring a transfer-function (TF) calibration. I investigated tube length, diameter, temperature, humidity, and other geometric and environmental factors to understand their influence on calibration consistency.

I owned the workflow from apparatus and test-matrix design through manufacturing, data acquisition, signal processing, statistical analysis, and presentation at the University of Toronto Undergraduate Engineering Research Day (UnERD). My MATLAB pipeline used FFT-based processing to calculate and compare transfer functions across test conditions.

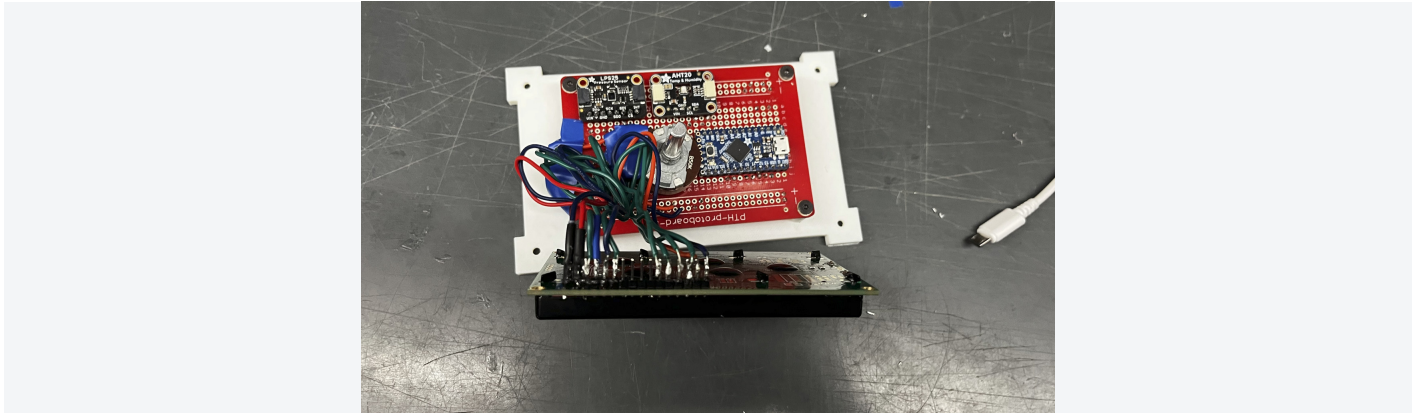
TOOLS USED

Design & Fabrication	Fusion 360 · SLA and FDM printing / slicing · Soldering and circuit design
Acquisition & Analysis	Arduino IDE · Python · MATLAB · DAQ and data collection · Signal processing and analysis

01 / UTIAS RESEARCH

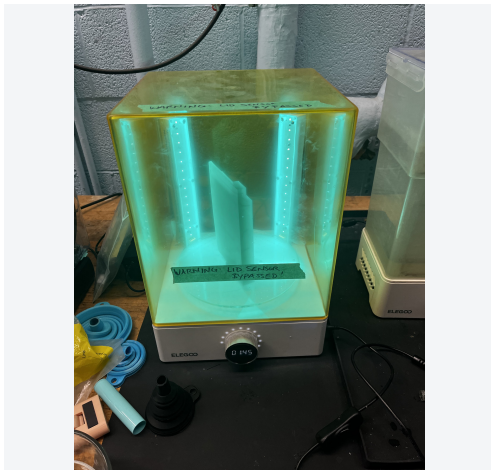
Instrumentation & Fabrication

I designed SLA-printed components, revised the calibration apparatus, and built an environmental sensor circuit using I2C and SPI. The FCET group retained the apparatus and circuit, alongside data and analysis tools, for subsequent calibration work.



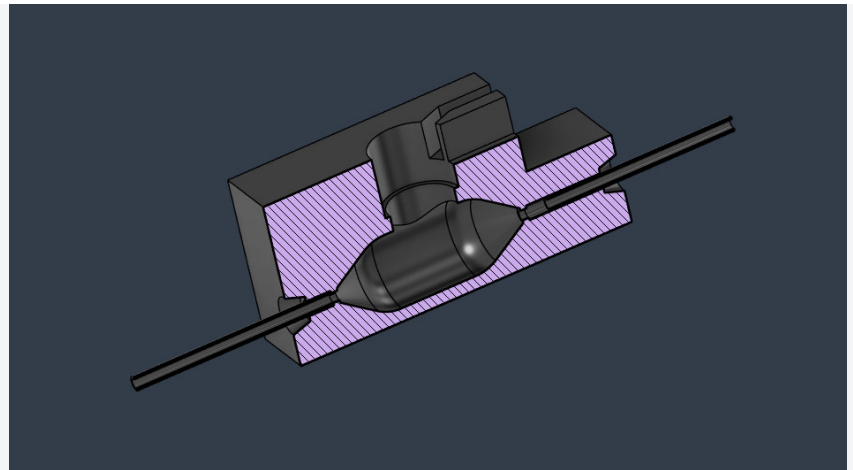
Sensor Circuit Internals

The sensor unit reads ambient temperature, humidity, and pressure using two sensors communicating over I2C and SPI. An Adafruit ItsyBitsy microcontroller, programmed in Arduino IDE, sends readings to an SPI LCD; a built-in potentiometer adjusts brightness. Sensor readings were manually compared with reference measurements for calibration.



SLA Printing and Resin Curing

Water-washable resin was selected for fine feature resolution and small layer height. Printed parts were coated with epoxy to seal pores and reduce the risk of acoustic leakage.



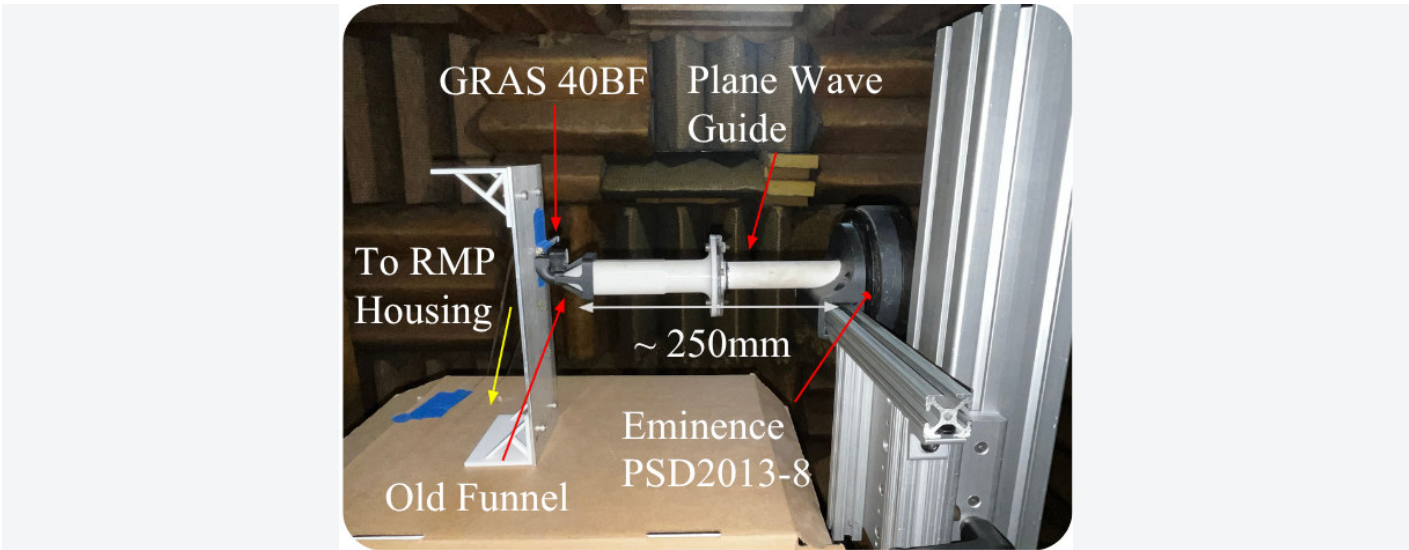
RMP Housing CAD

The microphone housing was measured from an existing print and redesigned in Fusion 360 to permit geometric modifications.

RELATING MEASUREMENTS TO PHYSICS

Interpreting the results required relating features in the transfer functions, including attenuation, phase shifts, resonant peaks, and other spectral artifacts, to the underlying acoustic and fluid behaviour within the pneumatic tubing. I compared these observations with analytical models of wave propagation, tube resonance, acoustic impedance, and viscothermal attenuation. This included examining how near-wall boundary layers and other tube-flow effects contributed to frequency-dependent pressure losses.

Calibration Outcomes



Calibration Apparatus in the Anechoic Chamber

The labelled setup records signals at 65,536 Hz with a 4,096-sample FFT window. MATLAB handles acquisition and transfer-function analysis.

Repeatability	69% lower TF magnitude standard deviation; 33% lower phase standard deviation relative to the control. Values are frequency-averaged across repeated calibrations.
Research Outputs	Mapped parameter effects to TF behaviour; proposed and experimentally evaluated improvements; delivered a retained sensor circuit, revised apparatus, and analysis tools; presented at UnERD.

Transfer-function smoothness was assessed using curve tortuosity: the ratio of normalized curve length to the straight-line distance between its endpoints. A value of one represents a straight line. Here, tortuosity describes the plotted transfer function, rather than the physical path of the tube. Repeatability was assessed separately using trial-to-trial standard deviation. See the full report for the methods and comparisons.

CHALLENGES + SOLUTIONS

Calibration Repeatability. Small changes in the relative positions of the microphone, speaker, and probe inlet introduced variability between calibrations. I redesigned the apparatus to hold their positions consistent, reducing transfer-function magnitude standard deviation by 69% relative to the control.

Interpreting Acoustic Behaviour. Changes in probe geometry produced complex shifts in attenuation, phase, and resonant features. I compared controlled parameter variations with wave-propagation and acoustic-impedance models to identify physical explanations and inform improved configurations.

Low-Frequency Coherence. The original speaker orientation produced poorer coherence at low frequencies. I revised its orientation relative to the reference microphone and probe inlet, improving coherence below 300 Hz.

[Full research paper \(PDF\)](#)

[UnERD poster \(PDF\)](#)

[MATLAB acquisition and analysis repository](#)

02 / RSX AERIAL DESIGN

CanSat Autonomous Glider

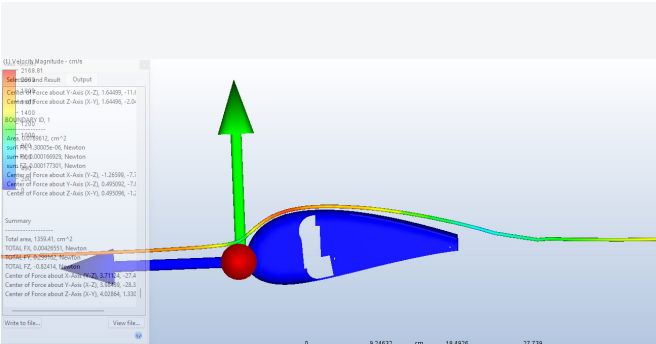
Aerodynamic Design | Mechanical Subsystem | 2025-2026 Competition

The Problem

Develop a compact, rocket-deployed autonomous glider and integrate its aerodynamic surfaces, structure, and mechanisms.

My Contributions

- Proposed the selected lifting-body fuselage and screened more than 20 airfoil profiles.
- Refined the fuselage using CFD, vortex generators, and endplates.
- Contributed to composite manufacture, tail components, and mechanical integration.



Glider Body CFD

Vortex generators and endplates address separation and spanwise flow losses in the lifting-body fuselage.

Approximately 5× simulated fuselage lift versus the baseline without flow-control devices at 20 m/s.

The fivefold result compares simulated fuselage configurations, not measured lift for the complete aircraft. Experimental confirmation remains a separate step.

DESIGN AND ANALYSIS WORKFLOW

Within RSX Aerial’s mechanical subsystem, I proposed the lifting-body fuselage concept selected for the final vehicle and led much of its aerodynamic development, from configuration studies through design refinement.

I screened more than 20 airfoil profiles and fuselage geometries using OpenVSP and VSPAERO, then selected promising configurations for Autodesk CFD analysis. Key flow trends and results were cross-checked using ANSYS Fluent. This staged process concentrated detailed analysis on the most promising candidates.

Pressure distributions, streamlines, wake development, and boundary-layer behaviour indicated premature separation and spanwise/tip-flow losses at low Reynolds numbers. Vortex generators and endplates delayed separation in simulation and increased fuselage lift approximately fivefold at 20 m/s.

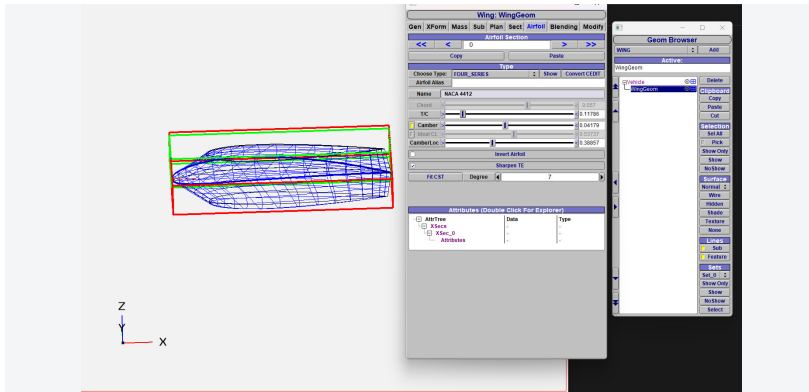
I also evaluated rocket nosecone geometries in Autodesk CFD and OpenRocket, supported CFD assessment of control mechanisms and aerodynamic interfaces, and assisted with composite wing manufacture. I refined parts for mechanical and electronic integration and manufactured and refined substantial portions of the tail and release mechanisms.

TOOLS USED

Design & Simulation	Fusion 360 · Autodesk CFD · ANSYS Fluent · OpenVSP / VSPAERO · OpenRocket
Build & Integration	Composite manufacturing · Mechanical and electrical integration

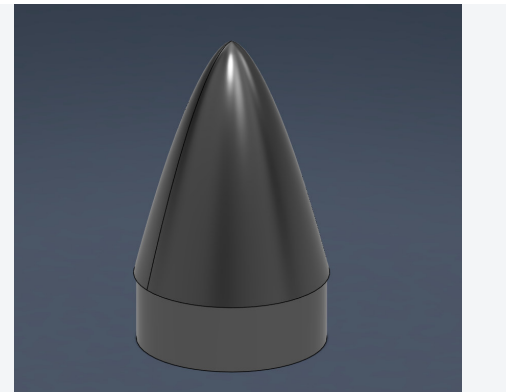
02 / RSX AERIAL DESIGN

Manufacturing & Integration



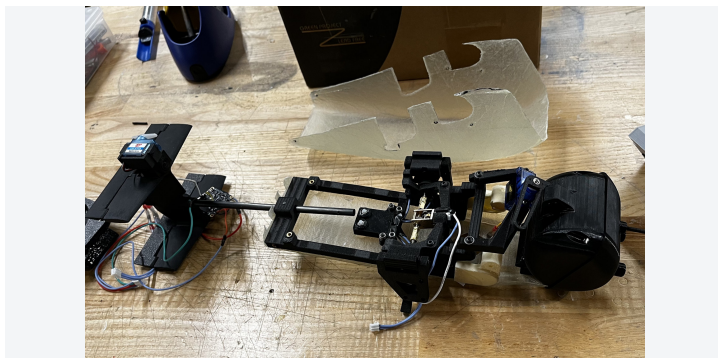
VSPAERO Screening

Profiles were compared at $Re \approx 97,000$ using lift coefficient versus angle of attack. Shortlisted profiles included NACA 4421, NACA 4424, and FX 69-PR-281.



CanSat Nosecone

Power-series nosecone geometries were screened in OpenRocket, then modelled and evaluated with Autodesk CFD to investigate drag.



T-Tail and Final Glider Manufacture

The final vehicle incorporates the fuselage concept I proposed. I contributed to fibreglass/resin composite manufacture, refined tail components for integration, and evaluated tail geometry in CFD. The printed frame used a nylon/carbon-fibre composite.

CHALLENGES + SOLUTIONS

Premature Flow Separation. CFD indicated separation and spanwise losses around the lifting-body fuselage. I incorporated vortex generators and endplates and compared the modified geometry with the baseline, obtaining approximately five times the simulated fuselage lift at the 20 m/s design speed.

Efficient Geometry Selection. Evaluating every candidate with detailed CFD would have required substantial iteration time. I used OpenVSP and VSPAERO to screen more than 20 airfoil profiles, then investigated the most promising configurations in Autodesk CFD and cross-checked key flow trends in ANSYS Fluent.

Manufacturing and Subsystem Integration. The T-tail and camera integration required reconciling CAD assemblies with manufacturing tolerances and the needs of both the mechanical and electrical subsystems. I refined component clearances and interfaces to support reliable fit and straightforward assembly. For load-bearing printed components, I selected build orientations with the expected loading and weaker interlayer bonds in mind, reducing reliance on interlayer adhesion in critical load paths.

TEAM COMPETITION RESULTS

Overall Score	70%
Canada	2nd
North America	8th

03 / INDEPENDENT EXPERIMENTAL RESEARCH

Flow Control Research Project

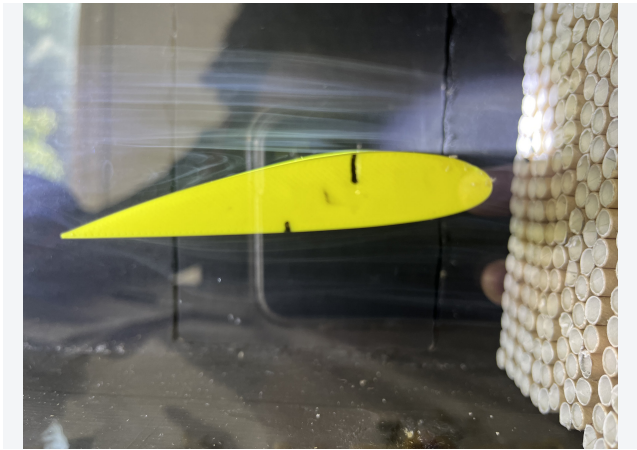
Independent Researcher and Builder | High-School Project | CWSF 2025

The Problem

Investigate how vortex-generator placement and spacing affect airfoil separation using a wind tunnel built at home.

What I Designed & Tested

- Built a smoke-visualization wind tunnel and eight 75 mm chord NACA 2412 test airfoils.
- Compared four chordwise positions and four gap spacings, with three trials per configuration.
- Used regression to identify predicted optima and compared them with the untreated airfoil.



Smoke Visualization
Separation and recirculation around a NACA 2412 airfoil. Incense smoke traces the flow over the wing.

76% higher predicted separation-criterion angle: 6.7° for the control versus 11.8° at the fitted optimum.

Visual criterion: separation reaching 30% chord. The fitted optimum is a prediction, not a force-measured stall angle.

APPARATUS AND EXPERIMENTAL METHOD

This independent project investigated vortex-generator placement and long-gap spacing on a NACA 2412 airfoil. Selected for its relevance to light aircraft such as the Cessna 172, the profile provided my first experience of aerodynamic research and passive flow-control testing.

Without formal laboratory access or dedicated aerospace mentorship, I built a low-speed smoke-visualization tunnel largely from cardboard and hot glue. At 0.84 m/s and 75 mm chord, the recalculated Reynolds number is approximately 4,300. I developed the apparatus, procedure, and analysis using the resources available.

I mathematically generated and 3D printed eight 75 mm chord airfoils. The test matrix compared four locations from 10% to 25% chord and four long-gap spacings from 2 to 5 mm. Separation reaching 30% chord defined the visual criterion as angle of attack increased; each configuration was tested in three trials.

Airfoil profiles were generated in Desmos and converted into STL files in Tinkercad. This was my first experience with CAD.

TOOLS USED

Design & Testing	CAD · Desmos · Tinkercad · Wind-tunnel testing and analysis
Methods	Experimental design, analysis and communication · Passive flow control

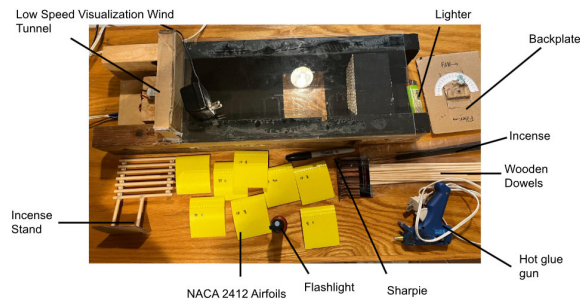
03 / INDEPENDENT EXPERIMENTAL RESEARCH

Experimental Findings



Custom Suction Fan

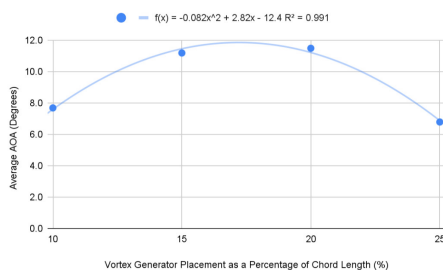
A 3D-modelled replacement inspired by vacuum-cleaner fans supplied the airflow required for visualization.



Labelled Apparatus

The cardboard-and-hot-glue tunnel, lighting, incense source, angle reference, and eight printed airfoils with varied vortex-generator configurations.

Average Boundary Layer Separation Angle as a Function of Vortex Generator Placement



Placement-Response Data

The experimental points and parabolic fit identify a predicted optimum in vortex-generator placement.

Placement had a stronger effect than gap spacing. Parabolic fits predicted optima of 17.2% chord and 3.6 mm spacing, with R^2 above 0.936. The fitted placement optimum raised the separation-criterion angle from 6.7° to 11.8° (approximately 76%). Some configurations performed worse than the untreated airfoil, demonstrating the need for deliberate optimization.

These are smoke-based separation measurements and regression predictions, not force-measured stall angles. The low-Reynolds-number experiment does not establish full-scale aircraft performance.

CYSF	Gold medal; second-best senior project; \$500 award. Selected as one of 15 projects from approximately 600 entries to advance to CWSF.
------	--

CWSF	Bronze medal among approximately 350 competing projects. Demonstrated passive-flow-control trends and airfoil flow visualization with limited resources.
------	--

CHALLENGES + SOLUTIONS

Insufficient Airflow. The initial fan did not generate enough airflow through the tunnel. I designed and 3D printed a replacement inspired by vacuum-cleaner suction fans, enabling smoke-visualization testing.

Defining a Repeatable Measurement. Without lift-measurement equipment, I needed a consistent way to compare airfoil configurations. I defined separation reaching 30% chord as the visual criterion and repeated each configuration across three trials.

Finding an Effective Configuration. Adding vortex generators did not consistently improve separation behaviour. I varied placement and gap spacing systematically and used regression to identify predicted optima, rather than assuming that any configuration would be beneficial.

[Full wind-tunnel report \(PDF\)](#)

[CWSF poster \(PDF\)](#)

04 / INDEPENDENT SOFTWARE DEVELOPMENT

Interactive 2D CFD Solver

Independent Developer | Ongoing Personal Project

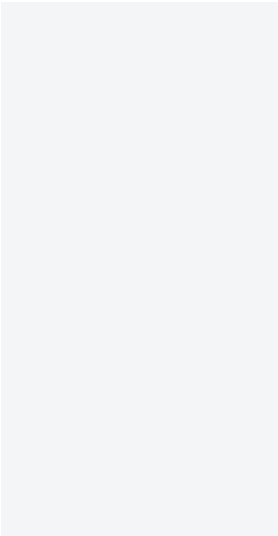
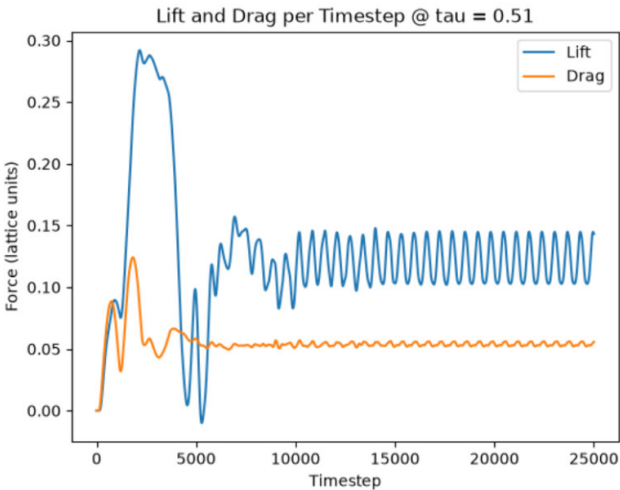
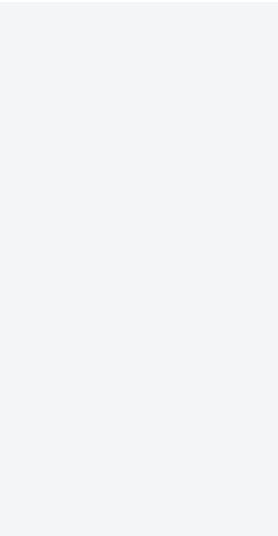
The Problem

Build an accessible CFD prototype that connects user-drawn geometry with simulated flow and aerodynamic force histories.

What I Developed

- Implemented a 2D LBM prototype with interactive geometry drawing and autofill.
- Refined boundary conditions and added force extraction.
- Compared NACA 0012 shedding frequency with a published numerical benchmark.

Approximately 1.8% difference in Strouhal number: 1.15 from the solver versus 1.1706 in the reference study.



Simulated Lift and Drag History

The original 25,000-step force history for NACA 0012 at 8° angle of attack and $Re = 2,000$. Lift oscillations become periodic after approximately 12,000 steps and provide the shedding-frequency estimate.

BENCHMARK COMPARISON

Geometry / Condition	NACA 0012 · Angle of attack 8° · Chord-based Reynolds number 2,000
Strouhal Number	Solver: 1.15 Published reference: 1.1706 Relative difference: approximately 1.8%
Definition / Scope	$St = f c / U_\infty$. The comparison assesses shedding frequency for one operating condition; it does not establish accuracy for all geometries or quantities.

Reference: D. F. Kurtulus - Vortex flow aerodynamics behind a symmetric airfoil at low angles of attack and Reynolds numbers (Figure 20 and Strouhal-number discussion)

04 / INDEPENDENT SOFTWARE DEVELOPMENT

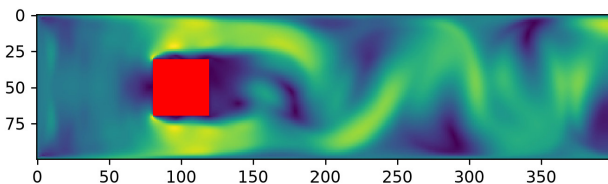
Solver Development & Evaluation

CURRENT PROTOTYPE

I am developing an interactive two-dimensional CFD tool using the lattice Boltzmann method (LBM), with the goal of making introductory flow simulation more accessible to beginners, hobbyists, and design teams.

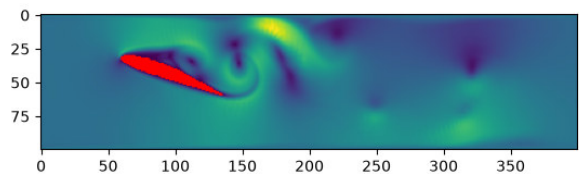
The working prototype allows users to draw a geometry on a grid and simulate flow around it. Implemented features include geometry autofill using breadth-first search, revised inlet and outlet boundary conditions, force extraction, and flow-field visualization. Development has also included collision-operator refinement and a subgrid-model implementation.

The prototype has been exercised with circular and square obstacles and NACA airfoils. A NACA 0012 benchmark compares the frequency of lift oscillations with a published numerical result. The current priority is extending quantitative checks of the solver; a simplified web/cloud interface remains a development goal.



Square-Obstacle Flow

Velocity is shown from blue (lower) to green/yellow (higher). Alternating von Kármán vortices form behind the square obstacle; the prototype maintains stability around its corners.



NACA 2412 Velocity Field

Flow visualization at 20° angle of attack shows separation and periodic trailing-edge vortex shedding.

BENCHMARK INTERPRETATION

For the benchmark shown here, $St = f c / U_\infty$, where f is the shedding frequency, c is chord length, and U_∞ is freestream velocity. The solver gives $St = 1.15$ at $Re = 2,000$ and an 8° angle of attack. Kurtulus reports $St = 1.1706$ for this geometry and operating condition using ANSYS Fluent, giving approximately 1.8% relative difference based on the displayed values.

The reference uses ANSYS Fluent. Agreement in shedding frequency supports this specific case; further grid, domain, and force-coefficient checks are needed to establish broader accuracy. Visual flow features are qualitative checks, not substitutes for quantitative evaluation.

CHALLENGES + SOLUTIONS

Numerical Stability. Maintaining stable simulations across different geometries required refinement of the collision operator and boundary conditions. I iterated on these components and tested the prototype with circular, square, and airfoil geometries.

Quantitative Benchmarking. Flow visualizations alone could not establish numerical accuracy. I implemented force extraction and used the periodic lift history to estimate shedding frequency, obtaining a Strouhal number approximately 1.8% from the published reference for the NACA 0012 test case.

[View the online project and simulation demonstrations](#)

Vortex Generators: Fluent CFD Study

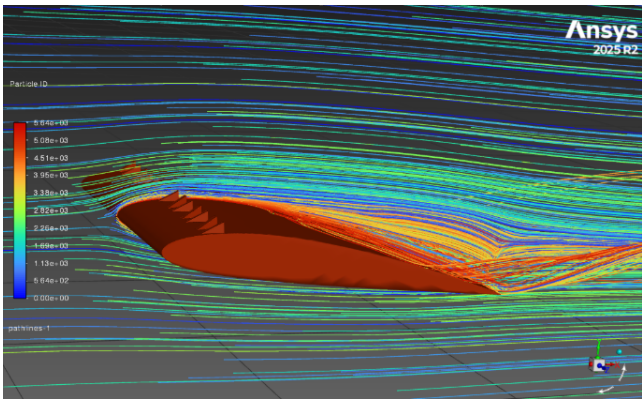
Independent CFD Investigation | Extension of the Flow Control Research Project | September 2026

The Question

Do control and VG configurations show similar separation locations in CFD at their experimentally identified separation-criterion angles?

What I Built & Checked

- Rebuilt the tunnel and airfoil in Fusion 360.
- Developed a steady laminar Fluent workflow.
- Compared meshes, force histories, mass balance, and signed wall shear.



Flow Around the VG Configuration

Pathlines show the 3D flow around the wing and VGs. Colours represent Particle ID, not speed.

Mesh sensitivity: 0.89% change in lift coefficient, 1.37% in drag coefficient, and a 1.3 mm shift in separation between the two VG meshes.

NUMERICAL SETUP

Geometry	NACA 2412: 75 mm chord × 65 mm span. VG elements: 2 mm high × 4.8 mm long × 1 mm thick.
Tunnel Domain	340 mm streamwise × 120 mm high × 75 mm wide; nominal 5 mm clearance at each wingtip.
Operating Point	Inlet 0.84 m/s along +X; airfoil rotated to each configuration’s experimental criterion angle.
Fluid & Model	ANSYS Fluent 2025 R2; steady, laminar, pressure-based. Air: $\rho = 1.225 \text{ kg/m}^3$; $\mu = 1.7894 \times 10^{-5} \text{ Pa}\cdot\text{s}$; $Re_c \approx 4,300$. Energy and gravity off.
Boundaries	Velocity inlet; pressure outlet at 0 Pa gauge; stationary no-slip wing, VG, and tunnel walls.
Numerical Methods	Coupled pressure-velocity scheme; least-squares cell-based gradients; second-order pressure and second-order upwind momentum; global pseudo-time.
Reference Values	Area 0.004875 m ² ; length 0.075 m; speed 0.84 m/s. Lift +Y; drag +X; forces integrated over airfoil/VG walls only.
Convergence Checks	Residual histories; unsmoothed lift/drag histories (Average Over = 1); inlet-outlet mass balance, targeting imbalance below 0.1%.

The large VG dimensions were retained to reproduce the original design. Low Reynolds number motivated the laminar baseline; the tunnel’s flow-conditioning filter supports a low-disturbance assumption but does not establish a measured turbulence intensity.

05 / FLUENT CFD EXTENSION

Mesh Sensitivity & Separation

For the “4 mm VG” case, geometry, angle, boundaries, and solver formulation were held fixed while surface/volume sizing and inflation-layer count were refined.

MESH SETTINGS

Setting	VG Medium	VG Fine	Far-Field Control
Global Minimum	0.21 mm	0.146 mm	0.25 mm
Global Maximum	8.5 mm	5.95 mm	10 mm
Local Curvature Minimum	0.21 mm	0.14 mm	0.25 mm
Local Curvature Maximum	5 mm	3.5 mm	5 mm
VG Face Target	0.20 mm	0.14 mm	Not applicable
Boundary Layers	22	25	18
Maximum Volume Length	8.5 mm	5.95 mm	10 mm

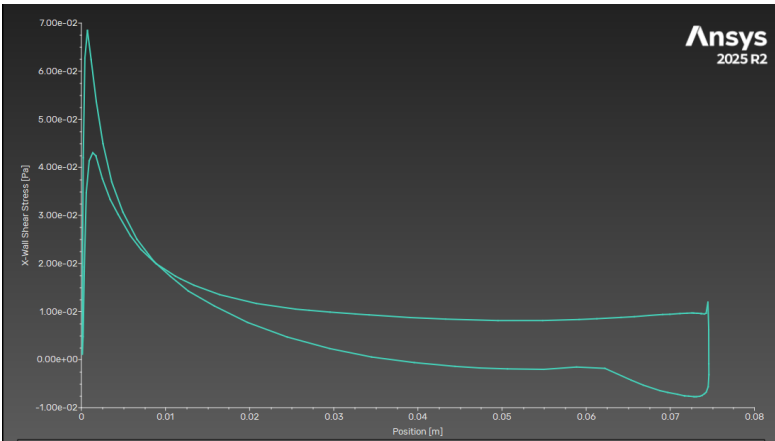
Polyhedral fill; Smooth Transition inflation on the airfoil/VG group, growth rate 1.2 and transition ratio 0.272. Tunnel walls remained no-slip without dedicated inflation after corner-quality problems; wall resolution remains a limitation. The far-field mesh was not separately refinement-tested.

VG REFINEMENT RESULTS

Quantity	Medium	Fine	Change
Lift Coefficient, C_L	0.4030	0.3994	-0.89%
Drag Coefficient, C_D	0.146	0.148	+1.37%
Separation Distance	34.5 mm	33.2 mm	1.3 mm upstream
Separation, x/c	46.0%	44.3%	-1.73 percentage points

These changes show limited sensitivity between the two tested meshes; they do not establish a formal discretization-error bound or mesh independence for every configuration.

LOCATING REVERSED NEAR-WALL FLOW



Signed X-wall shear identifies a zero crossing into reversed near-wall flow. Upper and lower surfaces form separate branches.

On a rotated, three-dimensional wing, this diagnostic also requires checking the local surface direction and flow field. Reported separation distances are referenced to the leading edge.

Findings & Tunnel Confinement

The original experiment used separation reaching 30% chord as its visual criterion. CFD revisited selected configurations at their corresponding experimental criterion angles.

Case	C_L	C_D	Separation	x/c
Control - Tunnel	0.390	0.125	36.9 mm	49.2%
Control - Far Field	0.1538	0.0768	33.6 mm	44.8%
4 mm VG - Medium	0.4030	0.146	34.5 mm	46.0%
4 mm VG - Fine	0.3994	0.148	33.2 mm	44.3%

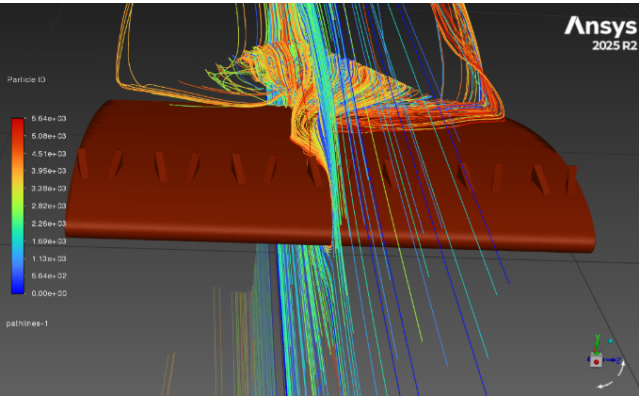
Control and VG coefficients are not a fixed-angle performance comparison. Percent chord is the reported leading-edge separation distance divided by the 75 mm chord.

Similar separation locations, shared offset: CFD placed separation at 44–49% chord versus the experimental 30% criterion. This gives preliminary relative consistency, with a downstream offset of 14–19 percentage points.

A shared offset suggests systematic error, including uncertain airspeed. It does not establish the cause: geometry, angle, inlet profile, wall resolution, visual criteria, and the steady-flow assumption remain possible contributors.

FAR-FIELD CONTROL COMPARISON

Domain clearances: $5c = 375$ mm upstream, $12c = 900$ mm downstream, and $5c$ on other sides. Inlet/outlet types were retained, with symmetry on distant lateral boundaries and a no-slip wing.



Domain Sensitivity

Compared with the tunnel control, the far-field run gave about 61% lower C_L , 39% lower C_D , and separation 3.3 mm earlier (4.4 percentage points of chord).

The contrast highlights sensitivity to confinement and boundary conditions. Mesh sizing and layer count also changed, so this is not an isolated wall-effect measurement. Separation location alone cannot explain the full force change.

Three-Dimensional Flow Structure

VG-wing pathlines complement wall-shear analysis. Colours show Particle ID, not turbulence or vortex strength.

ENGINEERING DECISIONS & NEXT CHECKS

Convergence checks combined residuals, force histories, and mass balance. Next: refine tunnel walls, test inlet-speed sensitivity, and run transient laminar analysis at 11.5° . The oscillating steady case is excluded from these findings.

[NASA grid-convergence guidance](#) | [NASA validation assessment](#) | [View CFD project online](#)

06 / INDEPENDENT DESIGN & FABRICATION

Tabletop Wind Tunnel

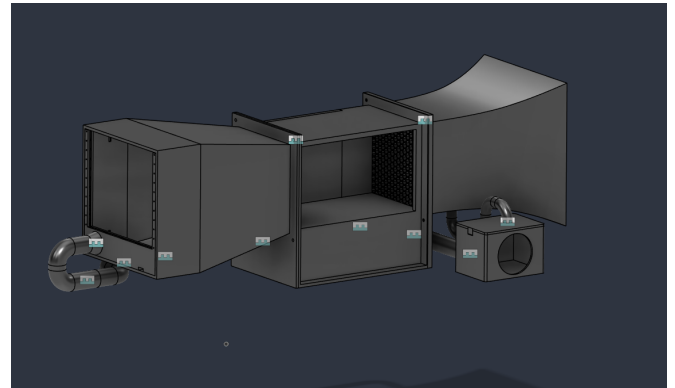
Independent Design and Fabrication | Ongoing Personal Project | 2026

The Problem

Build a compact smoke-visualization tunnel with a practical, modifiable structure and a reliable smoke-delivery path.

What I Developed

- Modular inlet, honeycomb, test section, and diffuser modelled in Fusion 360.
- PLA hardware fabricated on a Bambu Lab A1 mini, with iterative print preparation.
- A custom smoke chamber, injection rake, and fan-exhaust return circuit.



Integrated Tunnel Assembly

CAD layout of the inlet, honeycomb, viewing section, diffuser, and external smoke circuit. This view precedes the later cap and roof-height changes.

Visible smoke streaks achieved. The screw-on cap, switch to incense sticks, and taller chamber roof resolved the initial leakage and heat-distortion problems.

AIRFLOW AND MECHANICAL DESIGN

A Compact Suction Tunnel

A downstream fan draws air through a shaped inlet and honeycomb into the approximately 80 mm-high test section. The inlet contracts toward the viewing region; the honeycomb is intended to reduce transverse flow. A downstream diffuser expands the passage toward the fan.

Fan-Assisted Smoke Delivery

Part of the fan exhaust returns to the incense chamber to carry smoke toward the rake, revised to approximately 12 outlets of 1 mm diameter. The main tunnel remains open to the room: an open-circuit tunnel with a partial exhaust return for smoke delivery.

CAD TO PRINTED HARDWARE

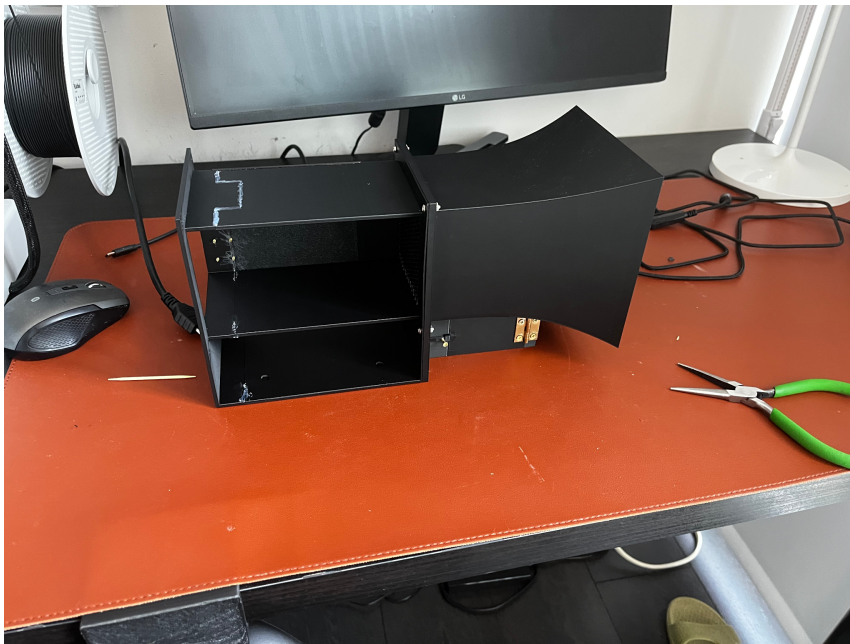
The CAD/CAM workflow combined Fusion 360 modelling, slicing, print preparation, and FDM fabrication. Nominal walls were reduced from 5 mm to 3 mm to limit material use. Modular sections allowed changes to the smoke route and interfaces without rebuilding the entire tunnel.

Tools: Fusion 360 / CAD · CAM / Print Preparation · FDM 3D Printing

Methods: Design for Manufacture · Flow Visualization · Iterative Testing

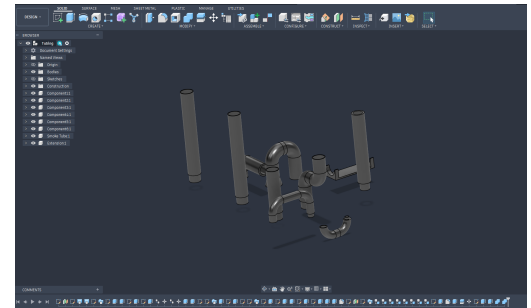
06 / INDEPENDENT DESIGN & FABRICATION

Fabrication & Iterative Testing



From CAD to the Workbench | Early Assembly

The curved inlet on the right joins the open test section on the left, with the honeycomb at their interface. Printed frames, fasteners, and the original hinged smoke chamber are visible. This assembly predates the screw-on cap and taller roof.



Smoke-Delivery Tubing

Straight sections, elbows, and connection features developed in Fusion 360 to route air and smoke around the tunnel. Iteration addressed smoke pooling and weak transport toward the rake.

Smoke Delivery in Operation

Visible streaks demonstrate smoke transport through the test section. The rake distributes the tracer while returned fan exhaust assists delivery. Visibility depends on smoke supply relative to the main airflow.

[Watch the demonstration online](#)

CHALLENGES + SOLUTIONS

Smoke Pooling and Weak Delivery. Smoke initially collected in the source chamber. Lid-opening tests exposed the influence of chamber airflow; revised routing and a fan-exhaust feed improved transport toward the rake.

Inconsistent Smoke Streaks. I investigated rake openings, gaps, filter resistance, and smoke dilution. Visible streaks were achieved during successful runs; consistency across repeated trials remains a development focus.

Incense Orientation and Supply. Vertically held sticks provided more useful smoke delivery than the earlier cone arrangement. Switching to sticks also reduced the intense local heat from cones.

Heat Distortion and Lid Leakage. A food-container screw-on cap resolved closure leakage. Switching to sticks and raising the chamber roof increased source clearance and resolved the observed heat distortion.

RESULTS AND NEXT STEPS

The prototype demonstrates printed-hardware integration and qualitative smoke visualization, with the initial sealing and heat problems resolved. Next steps focus on controlled source positioning, consistent fan conditions, and repeated trials. Airspeed, flow uniformity, and turbulence intensity have not been quantified.